

HILL AFB, UT, USA

WMO: 725755

Lat: 41.117N

Lon: 111.967W

Elev: 4789

StdP: 12.32

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 24101

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	9.4	13.0	0.1	6.6	14.8	4.5	8.3	18.4	28.0	26.6	24.8	30.5	10.8	100	0.611

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	24.0	94.3	60.8	91.8	60.1	89.7	59.7	64.5	84.9	63.3	84.4	62.1	83.6	7.1	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
58.2	86.6	71.5	55.3	77.8	72.2	53.3	72.3	72.1	32.1	84.5	31.0	84.6	30.1	83.7	73.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
24.2	20.6	18.8	DB	4.3	97.4	4.5	2.5	1.1	99.2	-1.5	100.6	-4.0	102.0	-7.3	103.8	
			WB	3.1	66.9	4.4	2.5	-0.1	68.7	-2.7	70.1	-5.1	71.5	-8.3	73.3	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	51.6	29.2	33.3	42.5	48.3	57.8	68.1	77.4	75.4	65.1	51.2	39.7	29.8
	DBStd	18.28	8.17	8.07	8.08	7.98	8.24	8.12	5.04	4.88	8.02	8.22	8.80	8.45
	HDD50	2562	646	469	257	126	23	2	0	0	6	84	322	627
	HDD65	5892	1111	888	696	503	249	62	2	4	94	431	760	1092
	CDD50	3142	0	1	25	75	266	545	850	788	458	122	12	0
	CDD65	996	0	0	0	2	27	154	387	327	96	3	0	0
	CDH74	10999	0	0	0	23	308	1798	4484	3451	898	37	0	0
	CDH80	4758	0	0	0	1	70	708	2214	1511	251	3	0	0
Wind	WSAvg	9.2	8.1	9.2	9.5	9.6	9.2	9.6	9.7	9.7	9.9	9.1	8.7	8.4
Precipitation	PrecAvg	17.1	1.3	1.3	1.7	2.1	1.9	1.3	0.6	0.7	1.5	1.6	1.6	1.4
	PrecMax	34.3	3.1	3.2	5.1	5.6	5.5	4.2	1.9	3.2	5.5	4.2	3.5	5.4
	PrecMin	8.3	0.0	0.1	0.1	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	5.0	0.8	0.9	1.2	1.3	1.5	1.2	0.6	0.8	1.5	1.1	0.9	1.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	50.3	57.2	69.8	77.1	86.4	95.2	98.7	95.1	89.4	78.8	64.1	53.9
		MCWB	41.1	41.7	48.7	52.4	57.3	60.5	62.3	61.4	60.0	54.3	46.4	42.9
	2%	DB	45.9	52.2	64.3	71.8	81.5	90.9	95.3	92.5	86.2	72.9	60.1	48.5
		MCWB	38.3	40.3	46.1	50.0	55.5	58.5	61.0	60.7	58.7	52.2	45.5	40.1
	5%	DB	42.9	48.2	60.1	67.0	77.2	87.7	93.2	90.5	82.7	68.4	56.5	44.9
		MCWB	36.7	39.1	44.3	48.1	53.9	58.0	60.7	59.9	57.2	50.6	43.6	38.0
	10%	DB	39.6	44.8	56.3	63.0	73.0	84.0	90.8	88.2	79.5	64.3	52.4	41.3
		MCWB	34.6	37.2	42.5	46.4	52.6	57.1	60.2	59.5	56.2	48.4	41.8	36.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	43.2	44.8	49.7	53.7	59.8	63.6	66.3	66.5	64.4	56.9	49.4	44.0
		MCDB	47.6	52.8	65.7	73.5	82.3	89.1	84.9	85.8	81.7	71.3	60.1	51.4
	2%	WB	39.4	41.8	47.4	51.2	56.8	61.0	64.9	64.3	61.4	54.4	46.5	41.2
		MCDB	43.8	49.8	62.6	68.1	77.0	85.5	85.2	84.1	79.7	67.9	57.0	47.6
	5%	WB	37.4	39.5	45.2	49.2	55.0	59.4	63.8	62.9	59.6	52.1	44.5	38.8
		MCDB	41.8	46.6	58.4	64.3	73.5	82.7	85.0	83.7	77.7	65.8	54.6	43.9
	10%	WB	35.3	37.6	43.3	47.4	53.3	58.1	62.5	61.6	57.7	49.9	42.5	36.6
		MCDB	39.6	43.5	55.0	61.1	70.3	80.4	84.3	82.9	74.6	63.4	51.7	40.8
Mean Daily Temperature Range	5% DB	MDBR	12.3	13.7	17.2	18.3	20.0	23.2	24.0	23.2	21.4	18.6	15.0	11.8
		MCDBR	14.4	16.5	22.0	24.7	25.4	27.2	26.5	25.4	25.0	23.4	19.2	14.2
		MCWBR	9.3	9.4	11.4	12.1	10.9	10.5	9.8	10.2	10.9	11.7	11.0	8.9
	5% WB	MCDBR	13.4	14.9	20.7	22.6	23.0	25.1	22.9	22.2	20.9	21.0	17.8	13.3
		MCWBR	9.4	9.1	11.0	11.7	10.6	10.7	9.1	9.4	9.5	10.6	10.8	8.7
Clear-Sky Solar Irradiance	taub	0.250	0.262	0.276	0.312	0.315	0.312	0.339	0.334	0.304	0.281	0.261	0.247	
	taud	2.469	2.506	2.554	2.450	2.495	2.541	2.483	2.490	2.563	2.608	2.594	2.516	
	Ebn at Noon	295	307	312	305	305	304	295	294	298	296	290	287	
	Edh at Noon	23	27	29	34	34	32	34	33	28	24	21	21	
All-Sky Solar Radiation	RadAvg	633	910	1369	1753	2068	2465	2389	2076	1682	1159	748	524	
	RadStd	65	99	93	108	170	174	105	97	114	79	75	66	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (9 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page